MILITARY SPECIFICATION

TUNGSTEN DELAY COMPOSITION

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for the procurement of tungsten delay composition. (See 6.1.)

1.2 Classification. The tungsten delay composition furnished under this specification shall be of five series, depending on the burning time range. Burning time will be specified by the procuring activity. (See 6.2.)

Series I	0.15 to 1.0 sec/in. at 25° centigrade (C)
Series II	1.0 to 5.0 sec/in. at 25° C
Series III	5.0 to 28.0 sec/in. at 25° C
Series IV	28.0 to 33.0 sec/in. at 25° C
Series V	33.0 to 38.0 sec/in. at 25° C

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-S-366	Sieves; Standard, Testing
PPP-B-621	Box; Wood, Cleated-Plywood
PPP-B-621	Box; Wood, Nailed and Lock Corner

FSC 1375

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tests. Be used for operating instructions of the test apparatus.

Military

MIL-P-116	Preservation, Methods of
MIL-P-217	Potassium Perchlorate
MIL-B-550	Barium Chromate
MIL-D-20550	Diatomaceous Earth
MIL-P-22264	Powders, Ignition, Gasless

Naval Ordnance Systems Command (Code Ident 10001)

OS 9765	Boron-Potassium Nitrate

Naval Air Systems Command (Code Ident 30003)

AS 2686	Tungsten Metal Powder, Pyrotechnic

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1234	Pyrotechnics: Sampling, Inspection and Testing

DRAWINGS

Naval Ordnance Systems Command (Code Ident 10001)

2249362	Test Element Loading Assembly
2249363	Delay Body
2249365	Output Charge Holder Loading Assembly
LD 518765	Test Element

PUBLICATIONS

Naval Ordnance Systems Command (Code Ident 10001)

OD 12051	Operating Instructions for Pyrotechni Delay Timing Apparatus

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 170-190	Hazardous Materials Regulations, Department of Transportation

(Copies are available from the Superintendent of Documents, Government Printing Office, Washington, D. C. Orders should cite "the latest issue and supplements thereto.")

3. REQUIREMENTS

3.1 Description. The tungsten delay compositions shall be homogeneous mixtures of tungsten powder, barium chromate, potassium perchlorate, and diatomaceous earth. Compositions shall be designated in series with corresponding burning time ranges in accordance with 1.2.

3.2 Materials. The materials used in the manufacture of the tungsten delay composition shall conform to the requirements of 3.2.1, 3.2.2, 3.2.3, and 3.2.4.

3.2.1 Tungsten powder. The tungsten powder shall conform to the requirements of AS 2686, type II. These tungsten particle size ranges are considered the most useful for the given burning time series. With proper adjustment of composition tungsten percentage, any particle size close to the suggested range may be used to achieve the desired burning time.

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Series	AS 2686, class	Average particle size, microns
I	1	0.5 to <0.9
II	2	0.9 to <2.5
III	3	2.5 to <5.0
IV	4	5.0 to <10.0
V	4 + 1,2	5.0 to <10.0 or blend of coarse (5.0 - 10.0) and small amount of fine (0.5 - 2.5) (10% - 25%)

3.2.2 Barium chromate. The barium chromate shall conform to MIL-B-550, class A, with the exceptions that the minimum purity shall be 97.5 percent, and 95.0 percent or more of the material shall pass through a U. S. No. 325 screen. The average particle size of the barium chromate shall be 5 microns or less as determined by 4.3.4.

3.2.3 Potassium perchlorate. The potassium perchlorate shall conform to MIL-P-217, grade A. In addition, the following percentages and granulations of potassium perchlorate are suggested for optimum performance:

Series	Burning time range, sec/in.	Percent $KClO_4$	$KClO_4$ granulation
I	0.15 to >1.0	5	F*
II	1.0 to >5.0	5	F*
IIIA	5.0 to >10.0	5	F*
IIIB	10.0 to >20.0	5	C**
IIIC	20.0 to >28.0	10	C**
IV	28.0 to >33.0	10	C**
V	33.0 to >38.0	10	C**

*3.0 to 8.0 micron FSSS
**-100+200 mesh USS sieve

3.2.4 Diatomaceous earth. For all series compositions, 5.0 percent diatomaceous earth shall be used. The diatomaceous earth shall meet the requirements of MIL-D-20550, grade B.

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3.3 Formulation. The material components forming the delay compositions shall be homogeneously mixed. Advisory manufacturing procedures are described in 6.4.

3.3.1 Composition. In order to meet the specific burning time characteristics required, the percentage composition by weight of the tungsten delay composition shall be determined by pilot mixture. (See figures 1 and 2 for effects of tungsten particle size and percentage tungsten.) Acceptance of the composition which conforms to the performance tests of 3.4, 3.5, and 3.6 shall be considered satisfactory compliance with the composition formulation. An advisory procedure for pilot mixture is given in 6.4.1 and 6.4.2.

3.4 Burning time at 24° ± 4° C. The burning time of tungsten delay compositions shall conform to the basic time and tolerance designed in the drawings, specification, or contract as furnished. The burning time of tungsten delay compositions is an average time ascertained under the provisions of 4.3.2. The percent coefficient of variation (C.V.) of observed burning times shall be not greater than indicated below for the ranges indicated.

Series	Burning time range, sec/in.	C.V., %
I	0.15 up to 1.0	7.0 (max.)
II	1.0 up to 5.0	5.0 (max.)
IIIA, IIIB	5.0 up to 20.0	4.0 (max.)
IIIC, IV, V	20.0 up to 38.0	3.0 (max.)

3.5 Low-temperature functioning. The burning time shall not be increased by more than 20.0 percent when the initial temperature is changed from 24° ± 4° C to -54° ± 2° C. The provisions of 4.4.3 shall be applicable in determination of low-temperature functioning. The percent coefficient of variation of observed burning time shall be as specified in 3.4 at all required temperatures.

3.6 Delay performance. There shall be no delay failures at room temperature or at low temperature. A delay failure shall be a condition where ignition powder ignites but the delay column does not burn through. Satisfactory performance of 4.3.2 and 4.3.3 tests with no delay failures shall constitute conformance to this requirement.

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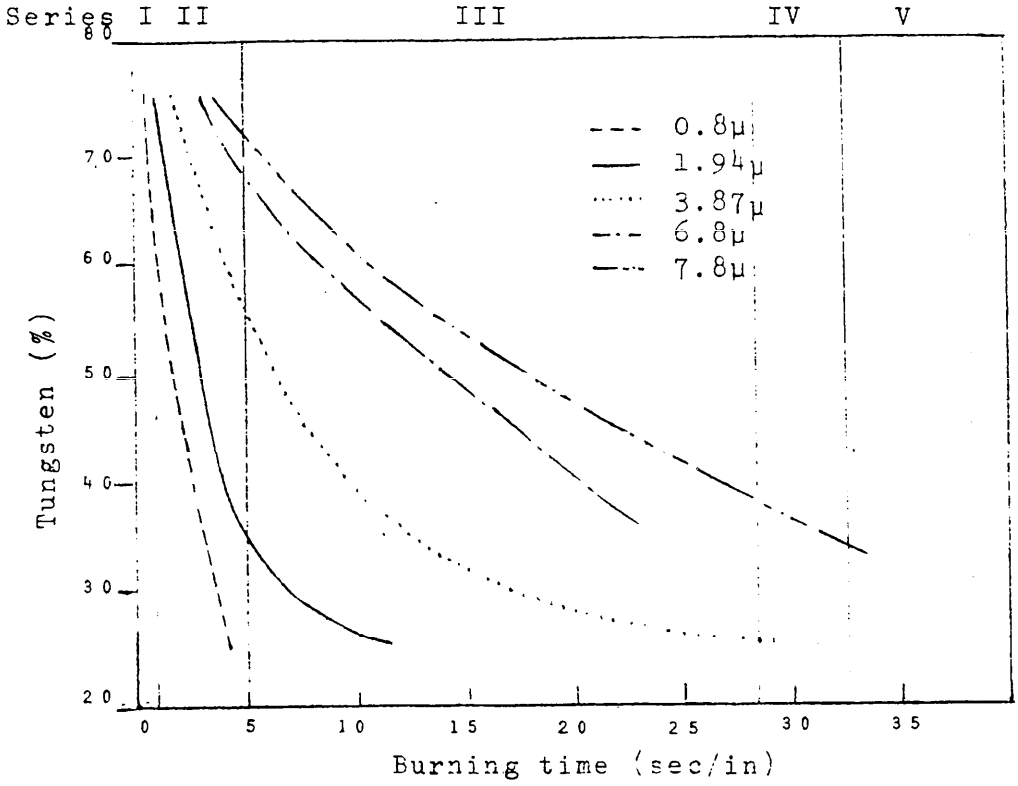


FIGURE 1. EFFECT OF THE PERCENTAGE OF TUNGSTEN ON BURNING TIME FOR FIVE TUNGSTEN PARTICLE SIZES

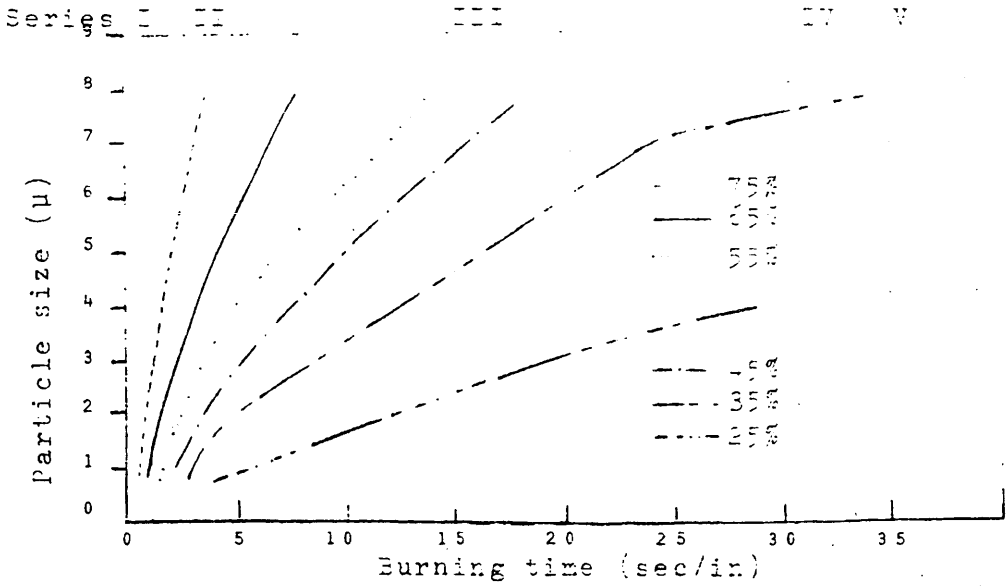


FIGURE 2. TUNGSTEN PARTICLE SIZE VERSUS BURNING TIME AT SIX DIFFERENT PERCENTAGES OF TUNGSTEN

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Lot. A lot of tungsten delay composition shall consist either of a manufacturer's batch which has been mixed as a unit from the ingredients or of a blend of one or more such batches.

4.1.2 Sampling. Sampling of the tungsten delay composition for test purposes shall be performed following the last mixing operation and screening of the dried powder. Each lot (4.1.1) to be tested shall be divided into 20 equal parts and 20 delay bodies shall be so loaded that each delay body contains a representative sample from one of the 20 parts without duplication (see 4.3.2). A second representative pooled sample of the delay composition is selected and a sufficient quantity to load a minimum of 10 test elements according to 4.3.2.1 for low-temperature functioning test (4.3.3). All sampling prior to test loading or loading in delay assemblies shall be carried out in areas having a relative humidity below 60 percent. The samples shall be stored in hermetically sealed containers and correctly labeled to show the following information:

(a) Name of item

(b) Weight of sample

(c) Lot number of delay composition

(d) Manufacturer

(e) Date of manufacture

(f) Lot number and manufacture of each material component used in the delay composition.

4.2 Acceptance tests. Unless otherwise specified, the contractor shall conduct acceptance tests specified (4.3.2 and 4.3.3) in the presence of the Government inspector to assure that the tungsten delay composition is in compliance with 3.4, 3.5, and 3.6.

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4.2.1 Test inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment facilities and personnel for performing all acceptance tests. The test equipment shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.3 Test procedures.

4.3.1 Ingredient testing. All ingredients shall be tested in accordance with the applicable specifications with amendments and the exceptions taken in section 3.

4.3.2 Burning time test for delay compositions. Conduct the test on the basis of each lot as provided in 4.1.1. The sample for test shall consist of 20 test elements as provided in 4.3.2.1. The lot shall be divided into 20 equal parts, and 20 delay bodies shall be loaded so that each delay body contains a representative sample from one of the 20 parts without duplication.

4.3.2.1 Loading of the test element. The delay body shown in Drawing 2249363 shall be loaded in accordance with Drawing 2249362. The tools required for loading appear in drawings under LD 518765. The ignition increment P shall consist of approximately 160 milligrams (mg) of A-1A ignition powder conforming to MIL-P-22264 or 160 mg of composition consisting of 80 percent A-1A ignition powder (MIL-P-22264) and 20 percent $BKNO_3$ ignition powder conforming to OS 9765, type II, class 3, and shall be adjusted to be within the drawing limits (0.060 ± 0.002 inch) for increment length when pressed at 30,000 ± 500 pounds per square inch. Each increment may be measured volumetrically by means of a scoop of such size as to give six approximately equal increments. After the last increment, D6 is pressed, the loading funnel cap shall be loosened and tightened again. The D6 increment shall then be re-pressed to improve compaction and to minimize the danger of loss of powder by fallout. The surface of the delay column shall be flush with the end of the delay element in which it is contained. Excess powder may be removed by sliding the end of the test element over a hard cloth. The output charge holder containing the P_2 charge shall be loaded in accordance with Drawing 2249365.

4.3.2.2 Timing of test elements. Suggested directions for mounting components into the delay body holder and assembling the test adapter are given in OD 12051. The temperature of the test elements for room temperature firing shall be within 24° ± 4° C. If detailed directions

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for testing at low and high temperatures are not specified for qualification of the powder, the following procedure shall be followed: The test adapters shall be conditioned continuously for at least 2 hours at the required temperature before firing.

4.3.2.3 Procedure for operating test apparatus. It is suggested that NAVORD OD 12051, Operating Procedure for Pyrotechnic Delay Timing Apparatus, be used for operating instructions of the test apparatus.

4.3.2.4 Computation of burning time and percent coefficient of variation. Compute the burning time which is the average (X) of the time observations obtained in 4.3.2.2. Compute the standard deviation (S) of the time observations. The symbols and the method of computation shall be governed by 6.3. Compute the percent coefficient of variation by dividing (100) × (S) by $\bar{X}$. Disposition of the lot shall be governed by 3.4 and 3.6. Where lot rejection is on the basis of the percent coefficient of variation, or 3.6, the lot may be remixed and resubmitted for acceptance under all provisions of 4.3.

4.3.2.5 Testing after shipment or prolonged storage. After tungsten delay composition has been properly mixed, it is possible for the fuel and oxidants to segregate under conditions of storage and transportation. Segregation appears to occur near the surface when a delay composition is subjected to prolonged agitation, particularly during shipment by rail. The surface layer is then nonhomogeneous and not representative of the composition of the lot. After shipment of a lot of tungsten delay composition or after storage of the lot longer than 3 months, the lot should be reevaluated according to test procedures outlined in 4.3.2. If the lot fails to meet the requirements of 3.4 then the lot shall be remixed to restore uniformity prior to test and use. In all testing, the provisions of 4.3.2 shall be followed to ensure representative sampling of the lot.

4.3.3 Low-temperature functioning test. Sampling shall be conducted after the last mixing operation (4.3.2.5). Select and prepare a representative pooled sample of delay mixture from the lot in accordance with 4.3.2 by preparing 10 test elements as provided by 4.3.2.1. Determine the burning time at low temperature by the procedures of 4.3.2.2 and 4.3.2.4 with the exception that the temperature at the time of firing and for at least 2.0 hours before firing shall be -54° ± 2° C. (A dry ice, forced draft cold box in which the draft is thermostatically controlled has been found satisfactory.) Determine average burning time according to 4.3.2.4. Compute the percentage change in burning time on the basis of change from the burning time determined under 4.3.2.4 at

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room temperature (24° ± 4° C). The lot is acceptable if the increase in burning time is less than 20.0 percent as required by 3.5 and meets the requirement of 3.6. A lot rejected for not conforming to 3.5 or 3.6 may be remixed and resubmitted for acceptance under 4.2. Failure of the lot to conform to 3.5 and 3.6 on resubmission shall result in rejection of the lot.

4.3.4 Granulation and average particle diameter determinations.

4.3.4.1 Granulation. Transfer an accurately weighed sample of approximately 10 grams to a dry, tared U. S. Standard No. 325 sieve, 3 inches in diameter and conforming to the requirements of RR-S-366. Wash the powder with a needle spray of water until no more powder passes through the sieve. Rinse the sieve with 10 milliliters anhydrous methanol. Drain well and dry 2 hours at 80° C. Cool in a desiccator and weigh the sieve plus any coarse material. Calculate the percentage passing through the sieve.

4.3.4.2 Determination of average particle diameter. Determine the average particle diameter by means of a subsieve sizer based on the air-permeability principle in accordance with MIL-STD-1234, method 202.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. In addition to the following, level A packaging shall be in accordance with the Code of Federal Regulations 49 CFR 170-190.

5.1.1.1 Unit packaging. Unit packaging of the delay composition shall be in accordance with MIL-P-116, method 1A5.

5.2 Packing. Packing shall be level A, B, or C as specified in the contract or purchase order.

5.2.1 Level A. In addition to the following, level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 170-190.

5.2.1.1 Exterior containers. The delay composition packaged as described in 5.1.1.1 shall be packed in an overseas-type wood box conforming to PPP-B-601 or PPP-B-621.

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5.2.2 Level B. In addition to the following, level B packing shall be in accordance with the Code of Federal Regulations 49 CFR 170-190.

5.2.2.1 Exterior containers. The delay composition packaged as described in 5.1.1.1 shall be packed in a domestic-type box conforming to PPP-B-601 or PPP-B-621.

5.2.3 Level C. The delay composition packaged in accordance with 5.1.1.1 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation and the Code of Federal Regulations 49 CFR 170-190.

5.3 Marking.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 170-190.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including lot number, burning time at room temperature and standard deviation, and percentage composition.

5.4 Data cards. Data cards shall be furnished with each lot; one to the procuring activity, one to the consignee, and one to be packed in each box. Each data card shall have the following information:

Tungsten Delay Composition

Manufacturer _____

Manufacturer's lot No. _____

Burning time (average) _____
(a) 24° ± 4° C _____
(b) -54° C _____

Standard deviation _____
(a) 24° ± 4° C _____
(b) -54° C _____

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Date of manufacture _____

Tungsten Powder: percent by weight _____

Manufacturer _____

Manufacturer's lot No. _____

Barium Chromate: percent by weight _____

Manufacturer _____

Manufacturer's lot No. _____

Potassium Perchlorate: percent by weight _____

Manufacturer _____

Manufacturer's lot No. _____

Diatomaceous Earth: percent by weight _____

Manufacturer _____

Manufacturer's lot No. _____

6. NOTES

6.1 Intended use. Tungsten delay composition is intended for use as a delay time determining element in ignition devices such as fuze explosive trains and cartridge actuated devices. The mixture burns substantially without evolution of gas; however, venting or expansion chamber pressure relief is desirable to relieve pressure generated by ignition elements. Application of this powder is restricted to hermetically sealed assemblies.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Burning time series number (see 1.2)

(c) Burning time

(d) Level of packing (see 5.2)

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(e) That the safety precaution requirements of the Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of Ammunition Ashore, OP 5, should be made applicable.

6.2.1 The attention of the contracting officer is invited to the quality assurance provisions and options in MIL-P-116 and to paragraph 5.3.2 of this document.

6.3 Computation of mean and standard deviation of a sample.

6.3.1 Take the sum of individual measurements (ΣX).

6.3.2 Compute the mean value ($\bar{X}$) by dividing the result of 6.3.1 by the number of measurements (N). Thus $\bar{X} = \Sigma X/N$.

6.3.3 Square each individual measurement (X^2).

6.3.4 Take the sum of the squared measurements (ΣX^2).

6.3.5 Multiply 6.3.4 by the number of measurements (N). This gives $N(\Sigma X^2)$.

6.3.6 Square the result of 6.3.1 $(\Sigma X)^2$.

6.3.7 Subtract 6.3.6 from 6.3.5. This gives $N\Sigma X^2 - (\Sigma X)^2$.

6.3.8 Multiply the number of measurements (N) by the number of measurements less one ($N - 1$).

6.3.9 Divide 6.3.7 by 6.3.8. This gives $[N\Sigma X^2 - (\Sigma X)^2]/[(N)(N - 1)]$.

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6.3.10 The standard deviation (S) is the square root of 6.3.9. Thus,

$$S = \sqrt{\frac{[N\Sigma X^2 - (\Sigma X)^2]}{N(N - 1)}}$$

6.4 Advisory procedures for formulation.

6.4.1 Pilot mixture. Since the particle size distribution of tungsten powder and the oxidants will vary slightly from lot to lot, the burning time of delay compositions made from different lots of ingredients may vary. A pilot mixture technique is therefore necessary to define the composition-burning time curve for the new components. A composition on the curve for the series of interest should be prepared (see 6.4.2) and its burning time determined according to paragraph 4.4.2. These data are used to construct a parallel curve on the appropriate graph as shown in figures 1 and 2 from which the composition for any desired burning time may be determined.

6.4.2 Processing pilot mixtures. Weigh the tungsten, barium chromate, and diatomaceous earth for a 100-gram batch to ±0.10 gram and place in a half-pint ball mill jar with 10 to 15 half-inch porcelain balls. Dry either in a vacuum oven, 20 ± 5 mm Hg, 71° ± 2° C for 2 hours, or in a vented forced draft oven for 2 hours at 71° ± 2° C. Weigh the potassium perchlorate into a separate container and dry simultaneously in the same oven. After drying, add the potassium perchlorate to the jar and cover immediately with a gasketed clamped lid before removal from the oven. Mill one-half hour minimum at 30 to 50 revolutions per minute. It is preferable to mill the mixture in an area with the relative humidity held below 50 percent. [Drying of the ingredients is necessary to avoid caking of the powder which retards and may prevent adequate mixing. A clear glass ball mill jar should be used so that the powder can be frequently inspected for centrifuging (from high milling speeds) or incipient caking. Persistent caking during mixing is preventable by more thorough drying of the ingredients and by reduction of the ambient relative humidity.] Screen the mixed powder through a U. S. Standard 40-mesh sieve.

6.4.3 Processing production batches. Dry the ingredients of the production batch at 71° ± 2° C until a sample weight of each shows less than 0.1 percent change at half-hour drying intervals. Before mixing, it is advisable that the ingredients be screened according to the following: tungsten, through U. S. Standard Sieve No. 325; barium chromate, through U. S. Standard Sieve No. 30; and diatomaceous earth, through U. S. Standard Sieve No. 20. Several mixing methods may be attempted.

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6.4.3.1 Dry mixing in a ball mill jar. A 300-gram batch of delay composition may be prepared by dry milling in a 1-quart ball mill jar with 0.5-inch porcelain balls. The charge and balls should not exceed one-half the free space in the jar. Follow the drying and operating procedure in 5.2 and mill between 1 to 2 hours. Screen the mixed powder through a U. S. Standard 40-mesh seive.

6.4.3.2 Dry mixing in a double-cone or other tumbling-type blender. Place the dried and weighed ingredients in the blender and seal with frictionless clamps. The batch should not fill more than one-third of the free space in the blender. Time of rotation will be determined by the blender design and may be expected to vary between 1.5 to 2.5 hours. Screen the mixed powder through a U. S. Standard 40-mesh sieve.

6.4.3.3 Dry mixing in a muller-type blender.

(a) Charge the dried, screened, and weighed ingredients into a muller-type blender such as the Simpson mixer. During mixing, operate the mixer by remote control. Stop the muller every 10 minutes and scrape the mixture from the plows, pan sides, and bottom into the active mixing area using a conductive plastic spatula. Minimum mix time shall be 30 minutes.

(b) After mixing, the delay composition may be pressed into blocks before screening to minimize component segregation during handling. This can be accomplished by spooning 2 heaping tablespoons (approximately 60 grams) of delay composition into a blocking die, such as the Carver 2-1/4-inch test cylinder outfit No. 22591, available from Will Scientific, Inc., and applying 8000 pounds minimum deadload for a dwell of 2 seconds with an air-hydraulic press such as the Denison Multipress. Care should be taken to spoon only that powder which can be removed without tilting or scraping the container. Otherwise, vibration induced component segregation may occur.

(c) Screen the processed powder through a U. S. Standard Sieve No. 20. Dry the powder either in an indirectly heated electric oven at 100° ± 5° C for 2 hours, or in a vented, forced-draft oven at 60° ± 10° C for 8 hours.

6.4.3.4 Wet mixing with a fluid shear mill.

(a) Premix the dried and weighed barium chromate and tungsten using a nonsparking container and spatula. Afterwards roughly blend this mixture with the potassium perchlorate and diatomaceous earth and charge it into a fluid shear mill such as the Abbe Dispersall slurry-type mixer, containing a measured amount of methanol (O-M-232, grade A).

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(b) The ratio of methanol to charge should follow approximately 0.45 ml of methanol per gram of solids for compositions covered by this document (30 to 40 sec/in.). This is an experimental ratio found necessary to give optimum slurry fluidity and mixing effectiveness. Compositions with burning times shorter than 30 sec/in. will have lower oxidant concentrations and the optimum fluid/solid ratio will become smaller. Mixing time will vary with size of the charge and capacity of the mill. After removal of the slurry from the mill, filter off excess methanol on a Buchner funnel, dry the powder at 71° ± 2° C under 20 mm Hg vacuum for a minimum of 12 hours or at 71° ± 2° C for 24 hours, in a well vented, forced-draft oven. Screen the powder through a U. S. Standard No. 40 sieve and package in hermetically sealed containers.

6.4.3.5 Wet mixing in a muller-type blender. Charge the dried and weighed ingredients into a 24-inch Simpson Mixer with a nonsparking spatula and operate the mixer dry for 5 minutes by remote control. Add ethyl alcohol (MIL-E-463B, grade 1) to the powder in sufficient quantity to form a viscous paste, after determining this amount experimentally with a small quantity of the powder. During mixing, turn the mixer on and off by remote control and keep the mixer completely isolated. Stop the mixer every 15 minutes to scrape down the sides and muller with a wooden spatula. Continue mixing until damp clumps of powder will crumble easily under pressure. Maximum mixing time will be 1.5 hours. Do not continue mixing until powder reaches an apparently dry state. On completion of the mixing operation, stop mixer and place receptacle under discharge hatch. Open hatch and operate mixer by remote control until powder ejection stops. Dry the powder at 71° ± 2° C under 20 ± 5 mm Hg vacuum for a minimum of 12 hours, or at 71° ± 2° C in a well vented, forced-draft oven for 24 hours. Screen through a U. S. Standard No. 40 sieve and package in hermetically sealed containers.

6.4.3.6 Judging mixing effectiveness and completeness. The completeness of the mixing operation can be measured by means of the quality assurance provisions and test requirements. The tungsten delay composition may be considered homogeneous if it meets the requirements of 3.4, 3.5, and 3.6.

6.4.3.7 Importance of drying after wet mixing. The drying directions for wet mixed tungsten delay compositions are critical. Volatile fluid residues not completely removed from the powder will cause excessive gassing and functioning failures. If the fluid residue contains appreciable moisture, physical segregation of potassium perchlorate may occur and cause malfunctions.

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6.4.4 Adjustment of the burning time of a production batch. The burning time of a production batch may differ from the pilot batch because of variation in mixing and preparative procedures between the two processes. Production batches prepared under nearly identical conditions from the same lots of ingredients may be expected to show nearly identical burning performance. If it is necessary to adjust the burning time of the production lot, this can be done in an analogous manner to the procedure under 6.4.1. After the new composition-burning-time curve is established the necessary amounts of each ingredient required to bring the mixture to a composition corresponding to the desired time is determined by calculation. Actual composition adjustments on the existing batch may then be done provided the adjusted batch meets all quality assurance provisions of this specification.

Custodian:
Navy - AS

Preparing activity:
Navy - OS

Project No. 1375- N102

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-T-23132A (AS), Tungsten Delay Composition

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
__ DIRECT GOVERNMENT CONTRACT __ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

__ YES __ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Attn: OR5
Indian Head, Maryland 20640

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